

Trainings ▾

Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Disconnect the batteries. Refer to Procedure 013-009 in Section 13. (</qs3/pubsys2/xml/en/procedures/99/99-013-009.html>)
- Disconnect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (</qs3/pubsys2/xml/en/procedures/102/102-012-022.html>)
- Remove the piston from the connecting rod. Refer to Procedure 001-054 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html>)
- Remove and discard the piston rings. Refer to Procedure 001-047 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-047-tr.html>)

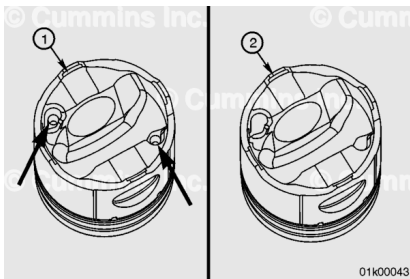
General Information

with Mechanically Actuated Injector

There are two types of pistons available for mechanically actuated engines. These are referred to as gallery crowned (1) and under crown cooled (2) pistons.

The gallery crown cooled piston (1) can be identified by two holes on the underside of the piston crown.

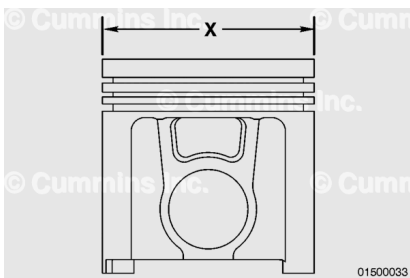
The under crown cooled piston (2) can be identified by no holes in the underside of the piston crown.



with Electronically Actuated Injector

There are two pistons available for electronically actuated engines. These are referred to as standard and cut back top land pistons. The standard piston has been in service since the release of the electronically actuated 38 and 50 liter engines and is currently used in most applications. The cut back top land piston is used only in QSK50 Power Generation engines built as of 12 June 2012.

The pistons can be identified by the difference in the piston top land diameter when measured as shown. The new piston has a smaller top land diameter than the old one.



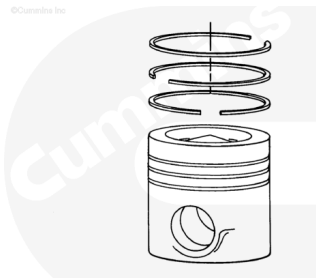
Piston Descripti on	Top Land Diameter (X)			
	Min		Max	
	mm	in	mm	in
Standard	156.96	6.18	157.06	6.183
Cut Back Top Land	155.69	6.13	155.79	6.133

Clean and Inspect for Reuse

with Mechanically Actuated Injector

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ CAUTION ⚠

Do not use the bead blast method to clean the piston. The piston will be damaged by blast material embedded in the aluminum, or the layer of anodized material will be removed.

Use a piston ring expander, Part Number ST-1269, or equivalent.

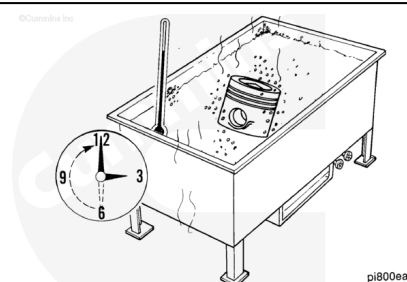
Remove and discard the piston rings.

Note : Premium pistons have an anodized surface that is gray in appearance. Do not remove this surface when cleaning.

Use steam to clean the outer layer of carbon from the piston surfaces.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



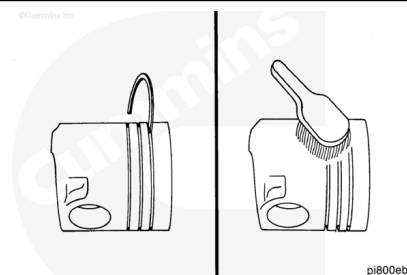
Use a kerosene emulsion based solvent. Do **not** use a solvent that is higher than 9.5 pH. Do **not** use a solvent that contains chlorinated hydrocarbons with cresols, phenols, or cresylic components.

Allow the pistons to soak in the solvent for a minimum of 30 minutes.

Note : For best results, soak the pistons several hours or overnight. Use a solvent that can be heated to 82°C to 93°C [180°F to 200°F]. Use a cleaning tank that will constantly mix and filter the solvent.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

⚠ CAUTION ⚠

Do not use a metal bristle brush. Damage to the piston ring grooves will result.

⚠ CAUTION ⚠

Do not damage the ring groove if you clean it with a used piston ring. The piston and ring can experience early failure.

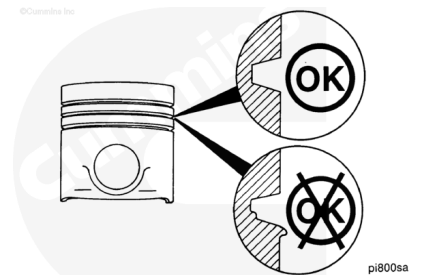
Use a brush or the gap end of a used piston ring to clean the ring grooves.

Repeat the soaking and scrubbing process until the piston is cleaned thoroughly.

Use steam to rinse the solvent from the piston.

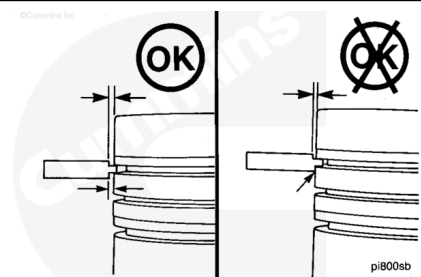
Dry with compressed air.

Check the ring grooves. The piston **must** be replaced if there is a visible ridge in the back of the groove. The piston **must** be replaced if a lip has formed on the outside diameter of the groove.



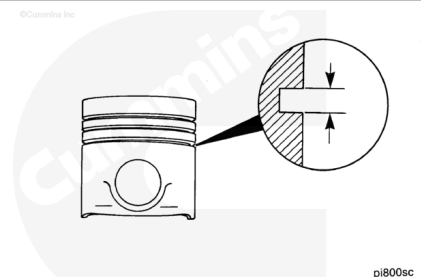
Use the piston ring groove wear gauge, Part Number ST-560.

Measure both compression ring grooves. The piston **must** be replaced if any measurement is **not** within specifications.



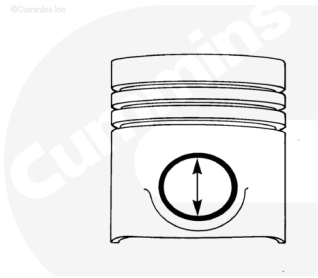
Measure the oil ring groove. The piston **must** be replaced if any measurement is **not** within specifications.

Piston Oil Ring Groove Width		
mm		in
4.788	MIN	0.1885
4.851	MAX	0.1910



The piston measurement is to be recorded at a room temperature of approximately 21°C [70°F].

Measure and record the inside diameter of the piston pin bore at several points as illustrated. The piston **must** be replaced when the smallest inside diameter measurement is **not** within specifications.



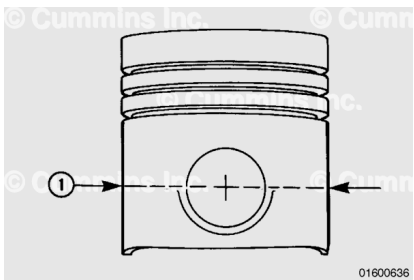
pi4brta

Piston Pin Bore Inside Diameter

mm		in
60.942	MIN	2.3993
60.952	MAX	2.3997

Check each piston for scratches or indications of scuffing. Reference Parts Reuse Guidelines, Bulletin 3810303, for details.

The piston measurement is to be recorded at room temperature of approximately 21°C [70°F].



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Piston Skirt Outside Diameter

	mm		in
Gallery Cooled	158.526	MIN	6.241
	158.558	MAX	6.242
Under Crown Cooled	158.552	MIN	6.242
	158.572	MAX	6.243

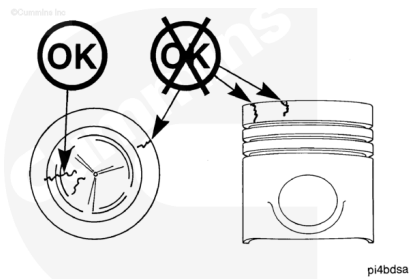
Note : Gallery cooled pistons are to be measured 25 mm [1.00 inch] from bottom of the piston skirt.

Note : Under crown cooled pistons are to be measured 50 mm [2.00 inch] from the bottom of the piston skirt.

Use the Crack Detection Kit, Part Number 3375432, or equivalent.

Note : The Crack Detection Kit contains the cleaner, penetrant, and developer necessary to check for cracks using the dye penetrant method.

Check for cracks on the top of the piston and in the piston pin bore.



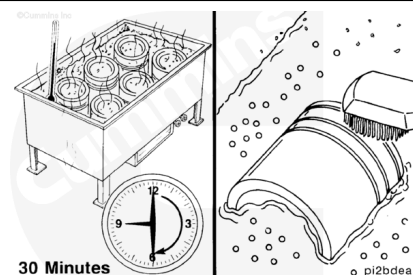
pi4bdsa

The piston **must** be replaced if there is a crack in the piston pin bore. All pistons except Part Number 3004730 **must** be replaced if there are any cracks on the top or the rim of the piston. This piston, Part Number 3004730, **must** be replaced if any crack extends more than 1/2 of the distance across the rim or if any crack on the top extends over the rim and toward the ring grooves.

with Electronically Actuated Injector

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ CAUTION ⚠

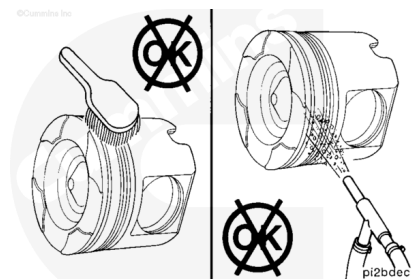
Be sure the cleaning solvent is approved for ductile iron. Damage to the pistons can result if an unapproved solution is used.

Allow the pistons to soak for a minimum of 30 minutes in a tank containing an approved cleaning solvent for ductile iron.

Use a hot, soapy solution and a non-metallic brush to remove carbon deposits.

⚠ CAUTION ⚠

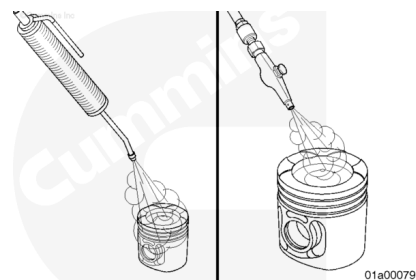
Do not use a metal brush. A metal brush will damage the piston ring grooves. Do not use glass beads to clean the grooves. Walnut shell or plastic bead, Part Number 3822735, blasting can be used on ring grooves and on the dome or crown of the piston. Use the minimum effective pressure and do not concentrate the spray in one area for an extended period. The recommended blast pressure for plastic bead blasting is 276 kPa [40 psi].



Clean the pistons with walnut shells or plastic beads.

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When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

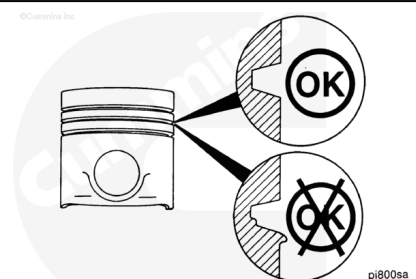


⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

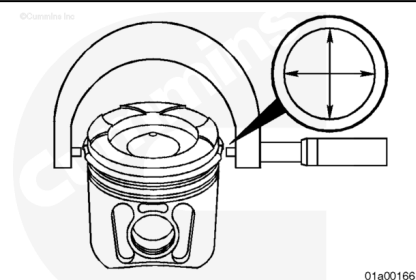
Use steam to clean the pistons and dry with compressed air.

Inspect the ring grooves. The piston **must** be replaced if there is a visible ridge in the back of the groove, or if a lip has formed on the outer diameter of the groove.



Use piston ring groove wear gauge, Part Number 3824846, and a 152.4 to 177.8 mm [6 to 7 in] micrometer to inspect the top groove (compression ring).

Measure the top and second ring grooves (diameter over pins) when the piston temperature is at 21°C [70°F] in two places, 90 degrees apart, as shown.



Ring Groove Wear Limits (Diameter Over Pins)

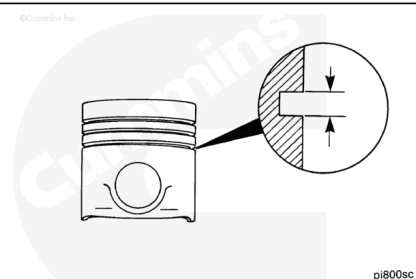
	mm		in
Top Ring	159.10	MIN	6.264
Second Ring	159.41	MIN	6.276

If the piston is **not** within specifications, the piston **must** be replaced.

Measure the oil ring groove.

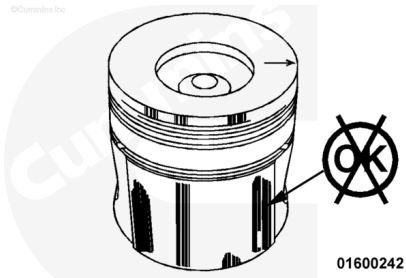
Piston Oil Ring Groove Width

	mm		in
Oil Ring	4.788	MIN	0.1885
	4.851	MAX	0.1910



If the piston is **not** within specifications, the piston **must** be replaced.

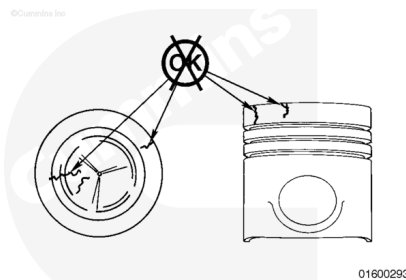
Inspect each piston for scratches or indications of scuffing. Refer to the Parts Reuse Guidelines, Bulletin 3810303.



Use crack detection kit, Part Number 3375432 or equivalent, to check for cracks on top and in the pin bore of the piston.

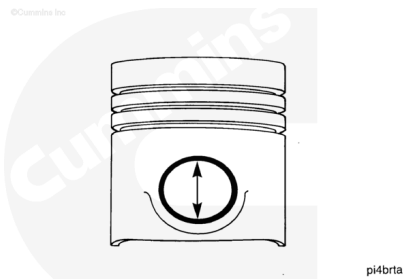
The piston **must** be replaced if there is a crack in the piston pin bore.

The piston **must** be replaced if there are any cracks on the top or the rim of the piston.



The piston measurement is to be recorded at a room temperature of approximately 21°C [70°F].

Measure and record the inside diameter of the piston pin bore at several points as illustrated. The piston **must** be replaced when the smallest inside diameter measurement is **not** within specifications.

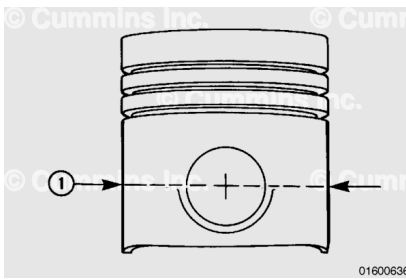


Piston Pin Bore Inside Diameter		
mm		in
60.97	MIN	2.400
60.99	MAX	2.401

If the piston pin bore is **not** within specifications, the piston **must** be replaced.

Measure the piston skirt at the centerline of the piston pin bore (1).

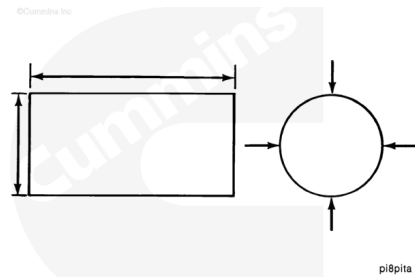
Piston Skirt Outside Diameter		
mm		in
158.631	MIN	6.245
158.671	MAX	6.247



If the piston is **not** within specifications, the piston **must** be replaced.

Measure the piston pin outside diameter and length.

Piston Pin Outside Diameter and Length			
	mm		in
Outside Diameter	60.929	MIN	2.3988
	60.935	MAX	2.3990
Length	118.550	MIN	4.667
	118.800	MAX	4.677



If the piston pin is **not** within specifications, the piston pin **must** be replaced.

Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.



- Install the piston rings. Refer to Procedure 001-047 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-047-tr.html)
- Install the piston in the connecting rod. Refer to Procedure 001-054 in Section 1. (/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)
- Connect the air/fuel supply to the air starter, if equipped. Refer to Procedure 012-022 in Section 12. (/qs3/pubsys2/xml/en/procedures/102/102-012-022.html)
- Connect the batteries. Refer to Procedure 013-009 in Section 13. (/qs3/pubsys2/xml/en/procedures/99/99-013-009.html)

- Operate the engine. Check for proper operation.

Last Modified: 11-Feb-2019
